Automate Cybersecurity Tasks with Python

Module 1

Programming: Used to create a specific set of instructions for a computer to execute tasks

Automation: The use of technology to reduce human and manual effort to perform common and repetitive tasks.

Syntax: rules that determine what is correctly structured in a computer language.

Comment: a note programmers make about the intention behind their code.

print() : Outputs a specificed object to the screen

Data type: Category for a particular type of data item.

String data: data consisting of an ordered sequence of characters

Float data: data consisting of number with a decimal point

Integer data: data consisting of a number that does not include a decimal point

Boolean data: data that con only be one of two calues: either True or False

List data: data structure that consists of a collection of data in sequential form

Tuple data: Data structure that consists of a collection of data that cannot be changed

Dictionary data: Data that consists of one or more key-value pairs. Each key is mapped to a value.

{ 1: “East”, 2: “West”, 3: “North”, 4: “South” }

Set data: Data that consists of an unordered collections of unique values.

Variable: A container that stores data

type() : Returns the data type of its input

Type error: An error that results from using the wrong data type

Conditional statement: A statement that evaluates code to determine if it meets a specifies set of condidtions.

if : starts a conditional statement

elif : else if

else : precedes a code section that only evaluates when all conditions that precede it within the conditional statement evaluate to False.

Operators:

> : greater than

< : less than

≥ : greater than or equal to

≤ : less than

== : evaluates whether two objects match

≠ : evaluates whether two objects are different

Logical operator for multiple conditions

and : if status ≥ 200 and status ≤ 226:

or :

not :

Iterative statement: Code that repeatedly executes a set of instructions, or loop

Type of loops:

for : Repeat a task for a specified sequence

while : Repeatedly execute based on a condidtion loop variable: A variable that used to control the iterations of a loop, i

range() : generates a sequence of numbers, range(0, 10) = 0 - 9

    Accepts 3 inputs: start point, stop point, increment by

    range(0, 5, 1)

    Start point is inclusive, meaning it will count from 0

    stop point is exclusice, meaning it won't count 5

break : break out of a loop

    if an if statement is true and you can break out of the loop

continue : skip an iteration and continue with the next one

Module 2

Function: a section of code that can be reused in a program

Built-in functions: functions that exist within python and can be called directly

    print()

    max() : Returns the largest numeric input passed into it

    min() : returns the smalled numeric inpuit passed into it

    sorted() : sorts the components of a list

user-defined functions: functions that programmers design for their specfic needs

def : placed before a function name to define a function

Parameter (Python): An object that is included in a function definition for use in that function

    def parameter():

Argument (Python): the data brough into a function when it is called

Return statement: A python statement that executes inside a function and send information back to the function call

Global varioable: A variable available through the entire problem.

    Outside functions

Local variables: variable assigned within a function can’t be called or accessed outside the body of function

Library: A collection of modules that provide code users can access in their programs

Module: A python file that contains additional functions, variables, classes, and any kind of runnable code

Python standard library: An extensive collection of usuable python code that often comes packaged with python.

re: useful to find patterns in log files

csv:

glob

os

time

datetime

Module 3

str() : converts the input object into a string

len() : Returns the number of elements in an object

string concatenation: the process of joining two string together

print(str + str)

Method: A function that belongs to a specfic data type

.upper() : Returns a copy of the string in all uppercase letters

.lower() : Returns a copy of the string in all lowercase letters

Index: A number assigned to every element in a sequence that indicates its position

Count indices from 0

.index() : Find the first occurrence of the input in a string and returns its location

Strings cant be immutable

immutable: cannot be changed after it is created and assigned a value

List concatenation: combining two lists into one by placing the elements of the second list directly after the elements of the first list

Lists can be immutable (changed or add to it)

.insert() : adds an element in a specific position inside a list

  x = which position we want to place

  y = what we want to insert

  .insert(x , y)

.remove() : Removes the first occurrence of a specific element in a list

Algorithm: A set of rules that solve a problem

.append() : adds input to the end of a list

Regular expression (regex): A sequence of characters that forms a pattern

+ : Represents one or more occurrences of a specific character

* : Zero one or more occurance

{n} : n occurance

{n,n} : min and max occurances

\w : matches with any alphanumeric character but it doesnt match symbols

\d : matches any single digit (0-9)

\s : matches any whitespace

. : any character

\. the literal period

ex: email = user1@email1.com

  \w+@\w+\.\w+

re.findall() : returns a list of matches to a regular expression

Module 4

Automate CI/CD with python:

Security testing:

- Static application security testing (SAST): Python scripts can be written to start SAST tools that look at your code weaknesses before it gets built.
- Dynamic application security testing (DAST): Python can run DAST tools to test software while its running in a test area.
- Software composition analysis (SCA): Python can work with SCA tools to check your software dependencies for weakness.

Automated Vulnerability scanner

Compliance checks

Secrets management automation

Policy enforcement

Python files

with : handles errors and manages external resources

- automatically close a file after reading it
- with open(”login_attempts.txt”, “r”) as file:
- r : read
- a : append
- w : write
- x : create

open() : open a file in python

.read() : converts files into string

parsing: the process of converting data into a more readable format

.split() : converts a string into a list

.join() : convert a list into a string

Types of errors:

- Syntax errors: Invalid usage of python language
- Logic errors: May not cause error messages, but can produce unintended results
- Exception: When program doesn’t know how to execute code even though there are no syntax errors